Chapter 8

Revision and consolidation

8.1 Revision strategy

Key ideas

Effective revision alternates retrieval from memory, explicit correction of errors and reuse in a problem. Passively rereading the lesson is not enough: you must attempt a question without model, compare your answer, then redo the question after a delay.

For each chapter, use the following code: **V** (mastered), **O** (fragile), **R** (to revisit). After the diagnostic, work first on an orange or red skill, then check immediately with a similar question.

Skill	V	O	R	Remediation action
Choose a model	☐	☐	☐	Repeat a exercise of sorting arithmetic-tic/geometric/exponential.
Read and calculate	☐	☐	☐	Revisit the targeted automatic skills.
Write and interpret	☐	☐	☐	Add data, formula, calculation, unit and a sentence of answer.

Session 52 — Diagnostic of prior knowledge

Revision and consolidation

11/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Identify what has been learnt.

· Classify the errors.

· Plan the remediation.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 25% increase. $1 + \frac{25}{100} = 1,25$

2. Give the formula for the general term of the arithmetic sequence from u_0. $u_n = u_0 + nr$

3. Solve $10^{x} = 1000$. $x = 3$

4. Differentiate $x + \frac{4}{x}$. $1 - \frac{4}{x^2}$

5. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: identify what is known and classify the mistakes.

Investigation

Answer 14 questions, two per chapter, without notes. Record each error in a grid: knowledge, choice of method, calculation, reading, writing. Then choose two personal priorities.

Detailed solutions · teacher edition

The correction awaits an error profile and a review plan, not only an overall score.
put into words the information that triggers the choice of method.

Written summary

The diagnosis is used to choose the remediation: recognise the error, recall the relevant rule, redo a similar question, then check without help.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The correction awaits an error profile and a review plan, not only an overall score.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed solutions · teacher edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

References in the textbook

Extensions: Revisiting chapters 1 to 7. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the textbook.

Calao textbook · p. 29, Exercise 52

In France, since 2016, sales of hybrid vehicles have risen sharply. The table below gives the number of hybrid vehicles sold in France per year.

Year	2016	2017	2018
Number of hybrid vehicles sold	7 420	11 010	15 350

Part A

1. Determine the percentage change between 2016 and 2017, then between 2017 and 2018.
2. Determine the geometric mean of 1.48 and 1.39.
3. What is the average annual rate of change between 2016 and 2018?

Part B

Assume that from 2018, sales of hybrid cars increase by 43% per year and that this trend will continue. We model it with the sequence (u_n) the change in sales where the part integer part of u_n represents the number of hybrid cars sold during the year $(2018+n)$.

1. Find u_0 ? Calculate u_1 and u_2.
2. What type of sequence is (u_n) ? Express u_n as a function of n.
3. According to this model, how many hybrid vehicles will be sold in 2023?

Detailed solutions · teacher edition

The rates are about 48 ,4% and 39,4%. The geometric mean of 1 ,48 and 1,39 equals $\sqrt{1{,}48 \times 1{,}39} \approx 1{,}434$, i.e. a mean rate of about 43 ,4%. In part B, $u_0 = 15\,350$, $u_n = 15\,350 \times 1{,}43^n$; for 2023, $n = 5$ and the integer part of u_5 gives about 91,788 vehicles.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 380 and the rate or common ratio by 6. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 53 — Revisiting: exponential functions

Revision and consolidation

13/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Revisit exponential models.

· Solve and interpret.

· Check the units.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 7 - x + 2$. $2x + 9$

2. Solve 10 $^{x} = 1000$. $x = 3$

3. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$

4. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

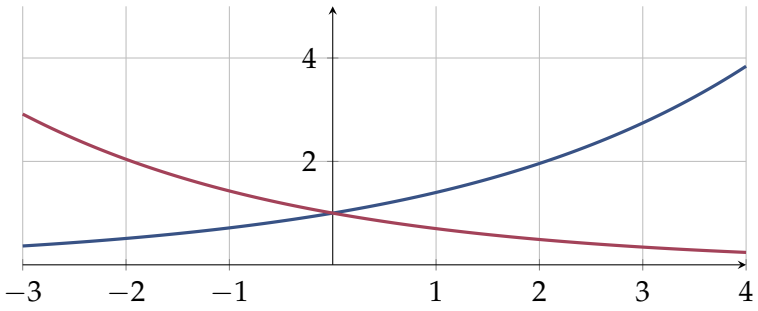
5. Give the formula for the general term of the arithmetic sequence from u_0. $u_n = u_0 + nr$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: revisit exponential models and solve and interpret.

Investigation

A value of 18,000 increases by 3 ,5% per year. Write the model, calculate the value after eight years and determine when it will exceed 25,000.



Detailed solutions · teacher edition

$V(t) = 18\,000 \times 1{,}035^{t}$, $V(8) \approx 23{,}703$. The threshold is exceeded from the tenth year on. Have students put into words the information that triggers the choice of method.

Written summary

An exponential function models a constant relative change. First identify the value initial value and the multiplier.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$V(t) = 18\,000 \times 1{,}035^t$, $V(8) \approx 23{,}703$. The threshold is exceeded from the tenth year on.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed solutions · teacher edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

References in the textbook

Extensions: Calao, p. 48, nos. 1 to 8, then p. 54, nos. 47 to 50. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 54, Exercise 58

The number of members of a basketball club has increased over the last three years by 2 ,5 %, then 4.1% and 3.8%. Calculate the average annual rate of change.

Detailed solutions · teacher edition

The overall coefficient is 1 ,025 × 1,041 × 1,038. The mean annual multiplier is its cube root, i.e. about 1.0346. The average annual rate of change is therefore about 3.46%.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with
Write in four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 54 — Revisiting: common logarithm

Revision and consolidation

18/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Revisit logarithms.

· Solve an equation.

· Check the conditions.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 160. 8
2. Differentiate $x + \frac{4}{x}$. $1 - \frac{4}{x^2}$
3. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
4. Give the formula for the general term of the arithmetic sequence from u_0. $u_n = u_0 + nr$
5. Solve $10^x = 1000$. $x = 3$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: revisit logarithms and solve an equation.

Investigation

Solve $10^x = 340$, $1{,}06^n = 2$ and $1{,}06^n \geq 2$. For each answer, distinguish the actual value, the value approximate value and smallest integer index.

Detailed solutions · teacher edition

$x \approx 2{,}5315$; $n \approx 11{,}90$; for the integer threshold, $n = 12$. Have students verbalise the information that triggers the choice of method.

Written summary

The logarithm turns an exponential equation into an equation in the exponent. Conditions: base positive and different from 1, right-hand side positive.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$x \approx 2{,}5315$; $n \approx 11.90$; for the integer threshold, $n = 12$.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed solutions · teacher edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

References in the textbook

Extensions: Calao, p. 70, nos. 1 to 10, then pp. 73-75, nos. 30 to 42. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 76, Exercise 43

Let x a strictly positive real number. The aim of the exercise is to determine an order of magnitude of x.

1. $x = 23.54$ and $y = 569{,}47$.
   a. The order of magnitude of a number is the power of 10 closest to that number. Give the order of magnitude of x and y.
   b. Using the calculator, determine $\log(x)$. Conjecture a link between $\log(x)$ and the order of magnitude of x. Check your conjecture with y.
2. $x = 5{,}4517$. Using a calculator, find the order of magnitude of x and $\log(x)$. The conjecture made in question 1 true?
3. Suppose that x is written in scientific notation: $x = a \times 10^n$ where $a \in [1; 10[$ and n is an integer relative. Give the order of magnitude of x depending on the value of a.
4. Deduce a method linking the order of magnitude of a strictly positive real number and its common logarithm.
5. Deduce the order of magnitude of the following two numbers: 2.26^{102} and 365.256^{2019}.

Detailed solutions · teacher edition

23,54 is closer to 10 than to 100, while 569,47 is closer to 1,000 than to 100; their orders of magnitude are 10 and 10^3. The change occurs at $a = \sqrt{10}$, not to $a = 5$. At scientific notation $a \times 10^n$, the order of magnitude is 10^n if $a < \sqrt{10}$ and 10^{n+1} if $a \geqslant \sqrt{10}$. We apply this rule to the fractional part of the logarithms: $102 \log(2{,}26) \approx 36{,}12$, so the order 10^{36} ; $2019 \log(365{,}256) \approx 5173.88$, so order 10^{5174}.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 390 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 55 — Revisiting: reciprocal function

Revision and consolidation

20/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Revisit the reciprocal function.

· Carry out a study.

· Draw and interpret.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 10% increase. $1+\frac{10}{100}=1,10$

2. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

3. Give the formula for the general term of the arithmetic sequence from u_0. $u_n = u_0 + nr$

4. Solve $10^{x} = 1000$. $x = 3$

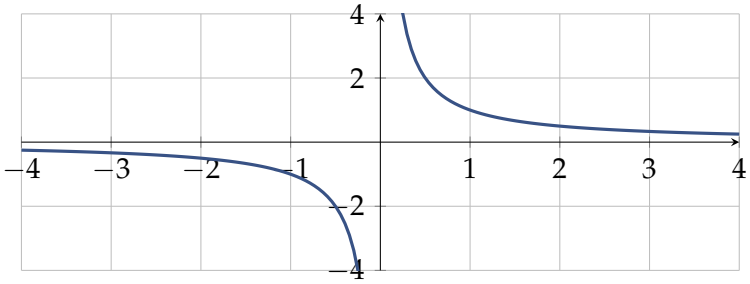
5. Differentiate $x+\frac{4}{x}$. $1-\frac{4}{x^2}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: revisit the reciprocal function and carry out a study.

Investigation

Study $f(x)=x+\frac{16}{x}$ on $]0,+\infty[$ and interpret its minimum in a context of average cost.
Produce the variation table and a sketch.



Detailed solutions · teacher edition

$f'(x)=\frac{(x-4)(x+4)}{x^2}$; minimum $f(4)=8$. Have students verbalise the information that triggers the choice of method.

Written summary

For a function containing $\frac{1}{x}$, first exclude $x=0$, then differentiate, factorise and study the sign before the variations.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed solutions · teacher edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

References in the textbook

Extensions: Calao, p. 94, nos. 1 to 9, then p. 96, nos. 10 to 21. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 102, Exercise 47

In Murano, a glassmaker calculated that the cost of production (in euros) of x vases is given by: $C(x) = x^2 + 30x + 400$. The craftsman wants to make as many vases as possible at a cost lowest possible average unit

Part A

1. Calculate $C(0)$ and deduce the craftsman's fixed costs. What is the production cost of 20 vases? For 20 vases, calculate the average cost of producing one vase.
2. Let $f(x)$ the average unit cost for x vases made. For a production of at least a vase ($x \geqslant 1$), express $f(x)$ as a function of x.
3. We assume that the average unit cost can be expressed by $f(x) = x + 30 + \dfrac{400}{x}$. Calculate $f'(x)$. Show that $f'(x) = \dfrac{(x-20)(x+20)}{x^2}$. Study the sign of $f'(x)$ and deduce the variations of f. How many vases must the craftsman produce for the average cost per unit to be minimal? Give that cost.

Detailed solutions · teacher edition

$C(0) = 400$ € : these are the fixed costs. $C(20) = 1\,400$ €, i.e. 70 € per vase. For $x > 0$, $f(x) = C(x)/x = x + 30 + 400/x$. Its derivative is $1 - 400/x^2 = ((x-20)(x+20))/x^2$. On $[1;+\infty[$, it is negative up to 20, then positive. The minimum is therefore reached for 20 vases and equals 70 €.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 395 and the rate or common ratio by 3. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

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